(ii) Compare the brightness of this image with that in (a). Explain. (2 marks)

Answers written in the margins will not be marked.

8. In the circuit shown in Figure 8.1, resistors R_1 and R_2 represent the heating elements in a heater using mains supply. Both resistors are immersed in water.

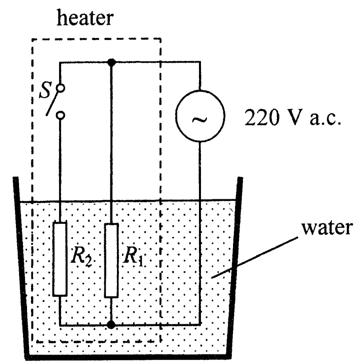

Figure 8.1

The heater can be operated in two modes, namely, heating and keeping warm, and it is controlled by the switch S. The power consumed by the heater in the heating mode is 550 W and in the mode of keeping warm is 88 W. The mains voltage is 220 V a.c.

(a) In which mode is the heater operating when switch S is open ? (1 mark)

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(b) Find the resistance of R_1. (2 marks)

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(c) When switch S is closed, calculate the current passing through resistor R_2. (3 marks)

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(a) (iii) Give **ONE** advantage of measuring the position of the second-order image instead of the first-order one. (1 mark)

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(b) In the experiment, the illuminated slit may not be well aligned along metre rule *A*. Suggest one way to reduce this error. (2 marks)

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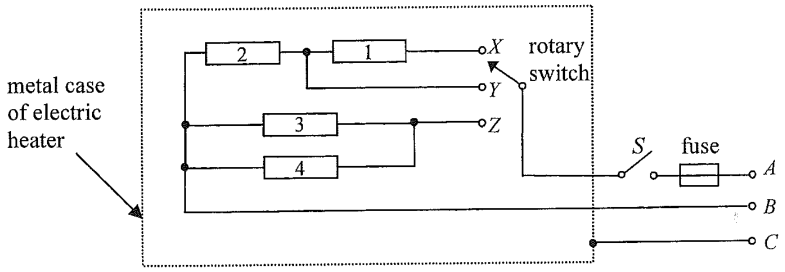
8. Figure 8.1 shows the schematic diagram of an electric heater consisting of four identical heating elements, each having a rated value of '500 W 220 V'. A user can use the rotary switch to select one of the three modes of operation X, Y, Z. Wires A, B, C from the heater are connected to the 220 V a.c. mains via a 3-pin plug.

Figure 8.1



(a) Find the resistance R of a heating element. (1 mark)

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(b) What is the total power dissipation when mode X is selected ? Assume that the resistance of each heating element remains unchanged. (2 marks)

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(c) Without the need of calculations, explain which mode of operation has the largest total power dissipation. (2 marks)

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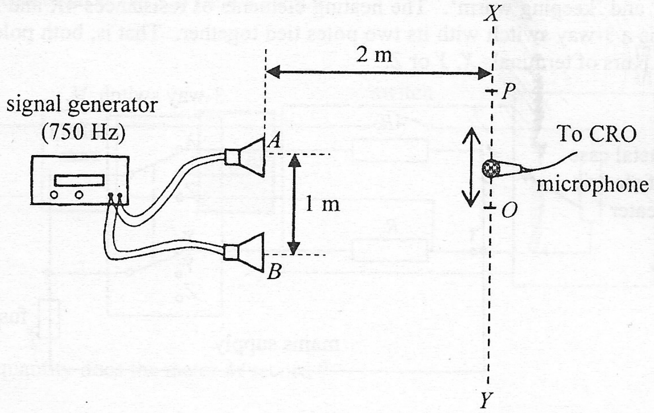
(b) To measure the speed of sound in air, a student connects two loudspeakers, A and B, to a signal generator as shown in Figure 7.2.

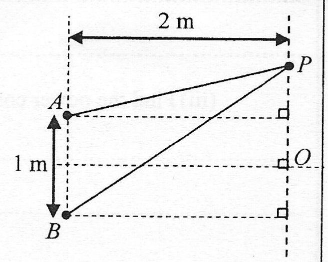
Figure 7.2



The separation of A and B is 1 m. A microphone is used to pick up the sound along the line XY at a distance of 2 m from the loudspeakers. The central maximum is at point O while the next maximum is at point P.

(i) With reference to the above settings, use the fringe separation equation $\Delta y = \frac{\lambda D}{a}$ in double-slit interference to find the wavelength λ of sound is not accurate. Explain briefly. (1 mark)

(ii) The distance between O and P is found to be 1 m when the signal generator is set at 750 Hz. By considering the path difference $PB - PA$, use the results of the experiment to find the speed of sound in air. (3 marks)



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8. (a) Figure 8.1 shows the schematic diagram of an electric heater which can operate in two modes, namely, 'heating' and 'keeping warm'. The heating elements of resistances $4R$ and R are connected to the mains supply via a 3-way switch with its two poles tied together. That is, both poles can be connected to one of the three pairs of terminals X, Y or Z.

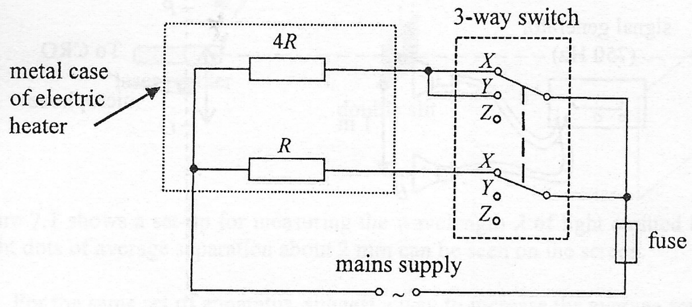

Figure 8.1

(i) To which pairs of terminals, X, Y or Z, should the switch connect to when the heater is in 'heating' mode ? (1 mark)

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The power consumed by the heater in 'heating' mode is 800 W.

(ii) Calculate the current drawn from the 220 V mains supply when the heater is in 'heating' mode. (2 marks)

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(iii) Find the power consumed by the heater in the mode of 'keeping warm'. (3 marks)

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8. Figure 8.1 shows a household electrical wiring circuit. The mains cable (containing live wire L and neutral wire N) is connected to a consumer unit via a kilowatt-hour meter *M*. At the consumer unit, the wires branch out into a number of parallel circuits.

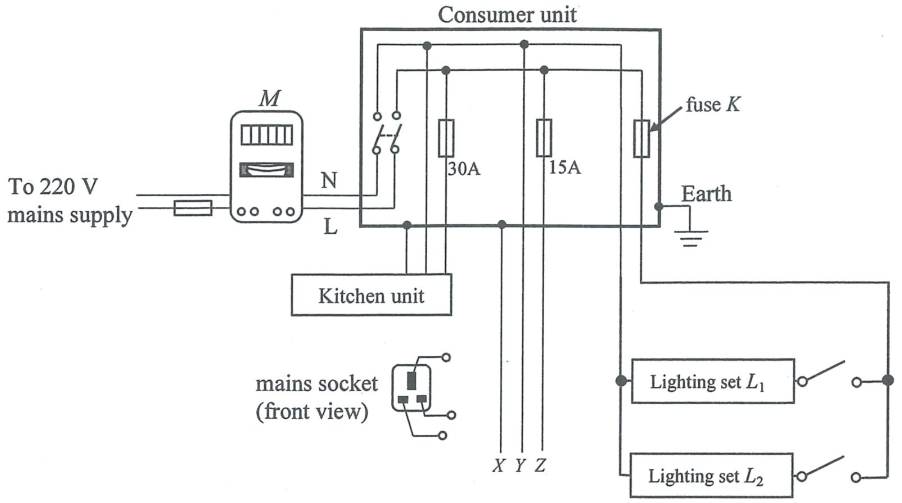

Figure 8.1

(a) Indicate on Figure 8.1 how the mains socket should be connected to wires X, Y and Z. (1 mark)

(b) Lighting sets L_1 and L_2 of power ratings 300 W and 450 W respectively are connected in parallel to the branch with fuse K.

(i) State one advantage of connecting L_1 and L_2 in parallel instead of in series to the branch. (1 mark)

(ii) If fuses marked 3 A, 5 A, 10 A and 13 A are available, which one is the most suitable to be fuse K ? Explain your choice. (3 marks)

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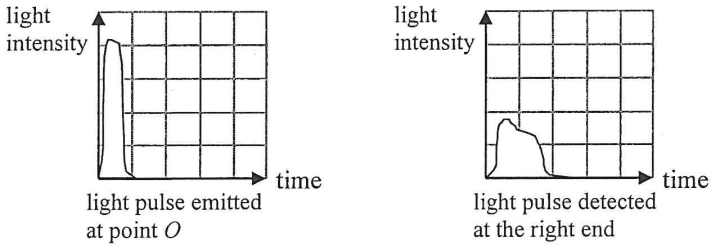
(b) A narrow monochromatic light pulse (i.e. of a short duration) emitted at point O propagates with its energy within $\theta = \pm 30°$ towards a light sensor located at the right end of the optical fibre. The respective emitted and detected light pulses are represented below using the same scales.

Figure 7.2

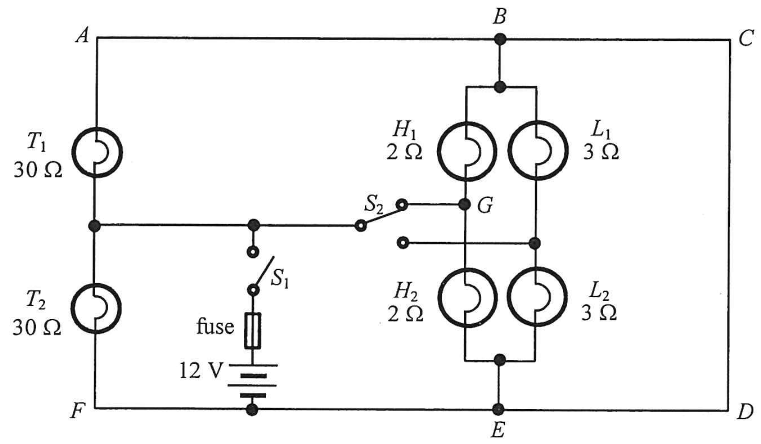


(i) Explain why the light pulse detected is **broader** (i.e. of a longer duration) and with **lower intensity**. Assume that the loss of energy of the light pulse due to absorption by glass is negligible. (2 marks)

(ii) An engineer suggests changing the refractive index n_c of the cladding in order to reduce the width of the light pulse received. Should n_c be increased or decreased ? Or, will a change in n_c have no effect on the pulse width ? Explain your choice. (2 marks)

8. Figure 8.1 shows a simplified circuit of the lighting system of a car. Each of the taillights (T_1, T_2), high-beam headlights (H_1, H_2) and low-beam headlights (L_1, L_2) has resistance 30 Ω, 2 Ω and 3 Ω respectively. The internal resistance of the 12 V battery and the resistance of the fuse are negligible.

Figure 8.1



When switch S_1 is closed and switch S_2 is set at the position shown in Figure 8.1, only T_1 and T_2 as well as H_1 and H_2 are lit. The current drawn from the battery is at a maximum in this setting.

(a) Explain why L_1 and L_2 are **not lit**. (1 mark)

(b) (i) What is the potential difference across the taillight T_2 ? (1 mark)

(ii) Indicate on Figure 8.1 the direction of current in each of the branches AB, GB and BC. Which branch carries the largest current ? (3 marks)

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7. Amy uses the motor of a toy fan as a simple generator. She connects a bulb to the two terminals of the motor. This is shown in Figure 7.1.

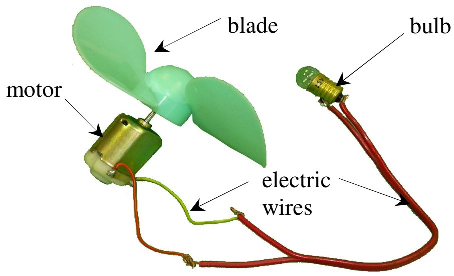

Figure 7.1

The bulb lights up when the blades are turned rapidly. Explain why and state the energy conversion taking place in this process. (4 marks)

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8. Figure 8.1 shows an earth leakage circuit breaker (漏電斷路器) installed in a domestic circuit. The live and the neutral wires pass through the centre of a soft iron ring of mean radius 1 cm. A 100-turn coil *C* with cross-section area 0.8 cm^2 is wound on the ring.

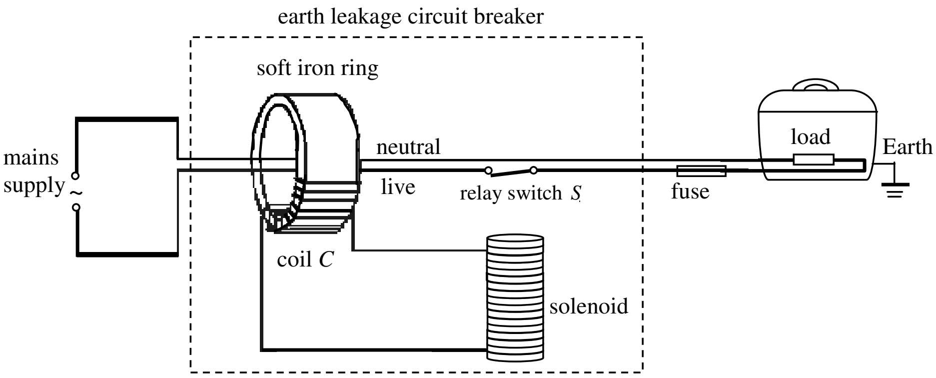

Figure 8.1

In case of an earth leakage in the domestic circuit such that the current in the neutral wire and the live wire differ by 0.5 A or more, the relay switch *S* opens and disconnects the mains supply. To reconnect the supply, *S* has to be reset manually.

(a) Explain why *S* opens when there is a leakage current of 0.5 A from the load to the Earth. (3 marks)